

Member ID: _____

Time: _____

Rank: _____



Computer Security

(320)

STATE 2024

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. Which encryption algorithm is commonly used for secure online transactions?
 - A. AES
 - B. RSA
 - C. MD5
 - D. DES
2. What is the purpose of a digital signature in cryptography?
 - A. Encrypting data for secure transmission
 - B. Authenticating the sender of a message
 - C. Preventing unauthorized access to data
 - D. Generating random numbers for encryption
3. Which authentication protocol provides secure communication over an untrusted network?
 - A. OAuth
 - B. SAML
 - C. Kerberos
 - D. RADIUS
4. Which access control model uses attributes and policies to determine access rights?
 - A. Mandatory access control (MAC)
 - B. Discretionary access control (DAC)
 - C. Role-based access control (RBAC)
 - D. Attribute-based access control (ABAC)
5. Which encryption mode provides confidentiality and integrity of data?
 - A. Electronic Codebook (ECB)
 - B. Cipher Block Chaining (CBC)
 - C. Counter Mode (CTR)
 - D. Output Feedback (OFB)
6. Which authentication protocol is commonly used for remote access authentication?
 - A. OAuth
 - B. SAML
 - C. Kerberos
 - D. RADIUS
7. What is the purpose of a DMZ (Demilitarized Zone) in network security?
 - A. To protect against DDoS attacks
 - B. To prevent social engineering attacks
 - C. To create secure VPN connections
 - D. To separate internal and external networks

8. Which of the following is an example of a network security protocol?
 - A. HTTPS
 - B. SNMP
 - C. SMTP
 - D. DNS
9. Which of the following is a common technique used to prevent brute-force attacks?
 - A. Implementing CAPTCHA
 - B. Enforcing password complexity rules
 - C. Using biometric authentication
 - D. Encrypting sensitive data
10. What is the purpose of penetration testing?
 - A. To identify vulnerabilities in a system
 - B. To encrypt sensitive data
 - C. To monitor network traffic
 - D. To block malicious websites
11. Which of the following is an example of two-factor authentication?
 - A. Username and password
 - B. Security question and answer
 - C. Smart card and PIN
 - D. Biometric fingerprint scan
12. What is the primary purpose of a firewall?
 - A. To prevent malware infections
 - B. To encrypt network traffic
 - C. To filter network traffic
 - D. To monitor system logs
13. Which of the following is an example of a secure coding practice?
 - A. Using hardcoded credentials in the code
 - B. Ignoring error handling and validation
 - C. Implementing input sanitization and validation
 - D. Storing sensitive data in plain text files
14. What is the purpose of regular vulnerability scanning?
 - A. To monitor network traffic for anomalies
 - B. To encrypt sensitive data
 - C. To identify and assess potential security vulnerabilities
 - D. To enforce strong password policies
15. What is the primary purpose of hardening internal Windows and Linux systems and services?
 - A. Enhancing system performance
 - B. Increasing network bandwidth
 - C. Strengthening security
 - D. Expanding storage capacity

16. Which of the following is a best practice for hardening internal Windows and Linux systems?
- A. Using default configurations
 - B. Disabling user authentication
 - C. Enabling auto-login for all accounts
 - D. Implementing strong password policies
17. Which of the following is a recommended practice for securing Windows and Linux systems against malware?
- A. Disabling antivirus software
 - B. Allowing the execution of unknown files
 - C. Regularly updating antivirus definitions
 - D. Running systems without a firewall
18. What is the purpose of implementing intrusion detection systems (IDS) in hardened Windows and Linux systems?
- A. To prevent all types of attacks
 - B. To provide real-time network monitoring
 - C. To disable network connectivity
 - D. To increase system vulnerability
19. Which of the following is a recommended practice for securing Windows and Linux systems against privilege escalation attacks?
- A. Granting full administrative privileges to all users
 - B. Disabling user account lockouts
 - C. Regularly monitoring system logs for suspicious activities
 - D. Using weak and easily guessable passwords
20. Which of the following is a recommended practice for securing Windows and Linux systems against network attacks?
- A. Running systems without a firewall
 - B. Disabling network encryption protocols
 - C. Implementing network segmentation and isolation
 - D. Allowing unrestricted access to all network services
21. Which of the following is a recommended practice for securing Windows and Linux systems against zero-day exploits?
- A. Disabling all system updates and patches
 - B. Using outdated and vulnerable software versions
 - C. Regularly applying vendor-provided security patches
 - D. Allowing anonymous access to network services
22. What is the purpose of conducting penetration testing on hardened Windows and Linux systems?
- A. To disable system notifications
 - B. To increase system vulnerability
 - C. To identify and exploit security weaknesses
 - D. To verify the effectiveness of security controls

23. Which layer of the TCP/IP model is responsible for establishing and terminating connections between hosts?
- A. Application
 - B. Transport
 - C. Internet
 - D. Network Interface
24. Which protocol is commonly used for transferring files over the internet?
- A. SMTP
 - B. FTP
 - C. ICMP
 - D. UDP
25. Which protocol is used to deliver email messages over the internet?
- A. SMTP
 - B. FTP
 - C. SNMP
 - D. HTTP
26. Which TCP/IP protocol is used to diagnose network connectivity issues?
- A. FTP
 - B. ICMP
 - C. DHCP
 - D. ARP
27. Which layer of the TCP/IP model is responsible for specifying the format of data exchanged between applications?
- A. Application
 - B. Transport
 - C. Network
 - D. Data Link
28. What is the first step in creating an effective security policy?
- A. Conducting a risk assessment
 - B. Identifying potential vulnerabilities
 - C. Implementing security controls
 - D. Updating system software
29. Which of the following is an example of a technical security policy?
- A. Social media usage policy
 - B. Password complexity policy
 - C. Employee termination policy
 - D. Physical access policy

30. What is the purpose of an incident response policy?
- A. Ensuring regular data backups
 - B. Identifying potential vulnerabilities
 - C. Outlining procedures for responding to security incidents
 - D. Defining password complexity requirements
31. Which security policy governs the use of mobile devices within an organization?
- A. Acceptable use policy
 - B. Wireless network security policy
 - C. Backup and recovery policy
 - D. Access control policy
32. In risk analysis, what does the term "exposure" refer to?
- A. The level of uncertainty associated with a risk
 - B. The potential impact of a risk event
 - C. The likelihood of a risk occurring
 - D. The combination of the probability and impact of a risk
33. Which wireless security protocol is vulnerable to offline dictionary attacks?
- A. WPA3
 - B. WPA2
 - C. WPA
 - D. WEP
34. What is the purpose of SSID hiding in wireless security?
- A. It prevents unauthorized devices from connecting to the network
 - B. It encrypts data transmissions between devices
 - C. It conceals the wireless network's name from being broadcasted
 - D. It blocks rogue access points from accessing the network
35. What is the purpose of a captive portal in wireless security?
- A. It authenticates users before granting access to the network
 - B. It encrypts data transmissions between devices
 - C. It detects rogue access points
 - D. It blocks unauthorized devices from connecting to the network
36. What is the purpose of WPS (Wi-Fi Protected Setup) in wireless security?
- A. It allows for easy and secure connection of devices to the network
 - B. It encrypts data transmissions between devices
 - C. It detects rogue access points
 - D. It blocks unauthorized devices from connecting to the network
37. Which of the following is a digital security certificate?
- A. SSL
 - B. VPN
 - C. Firewall
 - D. Router

38. What does CA stand for in the context of security certificates?
- A. Certificate Authority
 - B. Cryptographic Algorithm
 - C. Certified Access
 - D. Client Authentication
39. What is the purpose of a code signing certificate?
- A. To encrypt network traffic
 - B. To verify software integrity and authenticity
 - C. To secure email communication
 - D. To authenticate client devices
40. Which protocol is commonly used to transfer files securely over a network?
- A. FTP
 - B. HTTP
 - C. SSH
 - D. SMTP
41. Which type of certificate requires the highest level of verification and validation?
- A. Domain validation certificate
 - B. Organization validation certificate
 - C. Extended Validation certificate
 - D. Wildcard certificate
42. Which encryption key is used to encrypt data in an SSL/TLS connection?
- A. Public key
 - B. Private key
 - C. Session key
 - D. Root key
43. Which cryptographic hash function is commonly used in digital certificates?
- A. MD5
 - B. SHA-1
 - C. SHA-256
 - D. AES
44. Which certificate attribute confirms that a certificate has not been tampered with since it was issued?
- A. Serial number
 - B. Signature
 - C. Validity period
 - D. Thumbprint
45. What is the primary goal of security incident event management (SIEM) systems?
- A. Real-time monitoring and analysis of security events
 - B. Network traffic optimization
 - C. Anti-malware protection
 - D. Data backup and recovery

46. Which monitoring technique involves analyzing system logs and event records to identify security incidents?
- A. Network traffic analysis
 - B. Intrusion detection
 - C. Log analysis
 - D. Vulnerability scanning
47. Which of the following is an example of a security monitoring tool?
- A. Microsoft Excel
 - B. Wireshark
 - C. Google Chrome
 - D. Adobe Photoshop
48. What is the primary advantage of real-time security monitoring over periodic monitoring?
- A. Reduced false positive rate
 - B. Enhanced incident response capabilities
 - C. Lower resource utilization
 - D. Improved network performance
49. What is the purpose of security baselining in monitoring security infrastructure?
- A. To establish a performance benchmark for the network
 - B. To create backups of critical system files
 - C. To detect potential security incidents
 - D. To identify and remediate software vulnerabilities
50. What is the primary role of a security operations center (SOC) in monitoring security infrastructure?
- A. To oversee physical security measures
 - B. To analyze network traffic patterns
 - C. To detect and respond to security incidents
 - D. To manage user access privileges